SHUNTA IDE

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EDUCATION

Master of Engineering **Apr 2026 – Current**
Keio University, Tokyo, Japan

Bachelor of Engineering **Mar 2026**
Keio University, Tokyo, Japan

RESEARCH EXPERIENCE

Research Assistant **Apr 2026 – Current**
Shu Tanaka Laboratory, Keio University.

Undergraduate Researcher **Apr 2025 – Mar 2026**
Shu Tanaka Laboratory, Keio University.

- Research on quantum annealing, quantum optimization and combinatorial optimization
- Developed the new variable reduction method for constrained combinatorial optimization problems.
- Investigated the sampling of degenerate quantum system in quantum annealing

TEACHING EXPERIENCE

Teaching Assistant (Quantum Technology and Market Design) **Dec 2026**
Department of Economics, Keio University.

Teaching Assistant (Algorithm and Data Structure) **Oct 2026 – Nov 2026**
Department of Applied Physics and Physico-Informatics, Keio University.

Teaching Assistant (Statistical Mechanics) **Apr 2026 – June 2026**
Department of Applied Physics and Physico-Informatics, Keio University.

PUBLICATIONS

Proceeding

S. Ide and S. Tanaka “Unfair Sampling of Quantum Annealing in Weighted Graph Bipartitioning Problems,” 2026 IEEE International Conference on Quantum Computing and Engineering (QCE), Toronto, Ontario, Canada, to appear 2026); arXiv:2604.11449.

S. Ide, S. Kikuchi and S. Tanaka, "Multi-Penalty Sample Persistence Variable Reduction for Constrained Combinatorial Optimization Problems," 2025 IEEE International Conference on Quantum Computing and Engineering (QCE), Albuquerque, NM, USA, 2025, pp. 496-497, doi: 10.1109/QCE65121.2025.10413.

Submitted For Publication

S. Ide and S. Tanaka “Ground-State Sampling Correspondence between Penalty-Based and Constrained Quantum Annealing”, arXiv:2610.26687 (2026).

S. Ide, S. Kikcuhi and S. Tanaka “Extending Sample Persistence Variable Reduction for Constrained Combinatorial Optimization Problems”, arXiv: 2509.19280 (2025).

CONFERENCE PRESENTATIONS

Total: 4 (first author: 4)
International Conference (reviewed): 4(4)

Oral Presentation

S. Ide and S. Tanaka "Effective XY Interactions and Ground-State Sampling in Penalty-Based Quantum Annealing", International Network on Quantum Annealing Conference (INQA) 2026, Hotel Kompas Bled, Bled, Slovenia, November 2, 2026.

S.Ide and S. Tanaka "Unfair Sampling of Quantum Annealing in Weighted Graph Bipartitioning Problems", 2026 IEEE International Conference on Quantum Computing and Engineering (QCE), Metro Toronto Convention Centre Toronto, Ontario, Canada, September 17, 2026.

S. Ide and S. Tanaka "Improving Sampling Fairness in Quantum Annealing via Penalty Terms", Adiabatic Quantum Computing (AQC) 2026, Science Tokyo, Tokyo, Japan, June 25, 2025.

S. Ide, S. Kikcuhi and S. Tanaka "Extensions and Evaluation of the Sample Persistence Algorithm for Constrained Combinatorial Optimization Problems", International Network on Quantum Annealing Conference (INQA) 2025, Universitat Pompeu Fabra, Barcelona, Spain, November 3, 2025.

Poster Presentation

Total: 8 (first author: 7)
International Conference (reviewed): 1(1), Domestic Conference: 3(2), Workshop・Symposium: 4(4)

S. Ide and S. Tanaka "Unfair Sampling in Quantum Annealing for Constrained Combinatorial Optimization: Penalty Method and XY Driver", Quantum Innovation 2026, WINC AICHI, Nagoya, Japan, December 9, 2025.

S.Ide and S. Tanaka "Ground-State Sampling in Quantum Annealing for the Graph Bipartitioning Problem", The 38th Quantum Information Kanto Region Student Chapter, Tokyo, Japan, October 2, 2026.

S. Ide and S. Tanaka "Ground-State Sampling in Quantum Annealing for the Graph Bipartitioning Problem", The 7th SQAI General Meeting, Okinawa Institute of Science and Technology, Tokyo, Japan, September 28, 2026.

K. Oana, **S. Ide** and S. Tanaka "Effects of Finite-Time Quench on Solution Distributions in Quantum Annealing for Multi-Objective Optimization", The Physical Society of Japan 2026 Annual Meeting, University of Tokyo, Tokyo, Japan, September 16, 2026.

S.Ide and S. Tanaka "Unfair Sampling of Quantum Annealing in Constrained Combinatorial Optimization", The 37th Quantum Information Kanto Region Student Chapter, Yokohama, Japan, March 31, 2026.

S. Ide and S. Tanaka "Unfair Sampling of Quantum Annealing in Constrained Combinatorial Optimization", The 7th SQAI General Meeting, Okinawa Institute of Science and Technology, Okinawa, Japan, March 6, 2026.

S. Ide, S. Kikcuhi and S. Tanaka "Multi-Penalty Sample Persistence Variable Reduction for Constrained Combinatorial Optimization Problems", 2025 IEEE International Conference on Quantum Computing and Engineering (QCE), Albuquerque, New Mexico, USA, September 4, 2025.

S. Ide, S. Kikcuhi and S. Tanaka "Multi-Penalty Sample Persistence Variable Reduction for Constrained Combinatorial Optimization Problems", Quantum Innovation 2025, Congrès Square Grand Green, Osaka, Japan, June 29, 2025.

PROFESSIONAL EXPERIENCE

LayerX Inc., Tokyo, Japan **Feb – Sep 2025**
Software Engineering Internship

MoneyForward Inc., Tokyo, Japan **Sep 2024**
Software Engineering Internship (short-term)

LayerX Inc., Tokyo, Japan **Sep 2024**
Software Engineering Internship (short-term)

FastLabel Inc., Tokyo, Japan **Feb – Nov 2024**
Software Engineering Internship

Wanderlust, inc., Tokyo, Japan **Dec 2022 – Aug 2023**
Software Engineer, Co-founder

LEADERSHIP & ACTIVITIES

NICT NQC Quantum Camp Exploratory Human Resources Development Program **Sep 2025 – Feb2026**
Selected for a one-year research program organized by the National Institute of Information and Communications Technology (NICT), the Ministry of Internal Affairs and Communications, supporting self-directed research and development in quantum ICT through research funding and mentorship from academic and industry experts.

MITOU Target Program **June 2025 – Feb 2026**
Selected for MITOU Target Program (Software Development Utilizing Quantum Computing Technologies), Information-technology Promotion Agency, the Ministry of Economy, Trade and Industry (Project Leader).

Student Volunteer of Global Quantum Startup Event **May 2025**
Global Quantum Friendship Club, Organized by Quantum Forum, Japan

Host of the Cloud League & GDSC Keio University Hackathon event **Sep 2022**
Hosted a hackathon event for students to develop innovative solutions using cloud technologies. Supported by multiple Japanese technology companies including CyberAgent Inc., MoneyForward Inc., GMO Internet Inc., and more.

Community Manager of Cloud League (Engineering Community) **May 2022 – Mar 2023**
Engineer community consisting of Google Developers Student Clubs members in Keio University, University of Tokyo, Tohoku University, and Hitotsubashi University in Japan. Hosted events such as hackathons, workshops, and lectures supported by companies and Google.

Google Developers Student Clubs (GDSC) Keio University Member **Apr 2022 – Mar 2023**
Engineer community supported by Google in Keio University.

HONORS & AWARDS

Koizumi Shinzo Memorial Graduate Scholarship (fully funded, merit-based) **Apr 2026 – Mar 2027**
Awarded based on academic excellence and demonstrated potential as a future researcher, Keio University.

Best Poster Presentation Award **Mar 2026**
Quantum Information Kanto Region Student Chapter, Japan.

Department Chair's Award **Mar 2026**
Selected for the Department Chair's Award (outstanding graduation research and thesis award) in Department of Applied Physics and Physico-Informatics, Keio University.

ADDITIONAL INFORMATION

CERTIFICATES

TOEIC L&R 950 (L480 / R470) (Sep 2026) TOEFL iBT EXPIRED

SOSOCAI LIKS

Portfolio	profile.sprshunt.com	GitHub	github.com/shunta0213
LinkedIn	linkedin.com/in/sprshunt		

(Last updated at Sep 18, 2026)